Computational Physics Exercises

Week 08: Regression

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Exercise W08A Ordinary least squares fitting

In this exercise you will implement ordinary least squares (OLS) regression and use it to fit polynomial models. You will first use data sampled from a known parabola and then fit a double-well potential from a small set of sampled positions. Finally, you will investigate what happens when the polynomial model has as many, or more, parameters than there are data points.

A polynomial of degree p is written as

$$\hat{f}(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p = \boldsymbol{\phi}(x)^{\mathsf{T}} \boldsymbol{\beta},$$

where

$$\boldsymbol{\phi}(x) = \begin{bmatrix} 1 & x & \cdots & x^p \end{bmatrix}^{\mathsf{T}}.$$

For sampled positions x_i, place the corresponding feature vectors in the rows of the design matrix $\mathbf{X}_{\text{aug}}$. The OLS coefficients are then obtained from the normal equations,

$$\boldsymbol{\beta} = \left(\mathbf{X}_{\text{aug}}^{\mathsf{T}} \mathbf{X}_{\text{aug}} \right)^{-1} \mathbf{X}_{\text{aug}}^{\mathsf{T}} \boldsymbol{y}.$$

Task 1: sample a quadratic function

Consider the function

$$f(x) = \frac{4}{5} \left(x + \frac{1}{4} \right) \left(x - \frac{5}{2} \right).$$

Sample the function at

$$x \in \{-2, 0, 1, 2, 4\}.$$

Plot the function over a suitable interval and add the five sampled points to the plot.

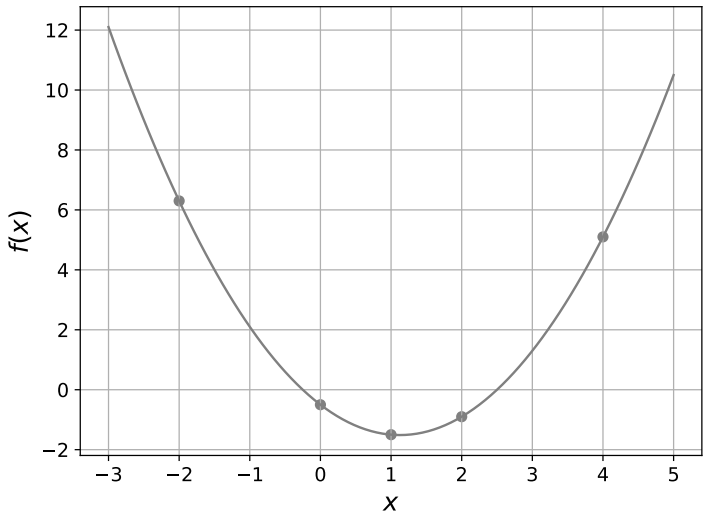


Figure 1: The quadratic function and the five sampled data points.

Task 2: implement polynomial OLS fitting

Implement polynomial OLS fitting in a class with `fit`, `predict`, and `plot` methods. The polynomial degree should be supplied when the class is instantiated. You may use the following outline:

```
class OLSFitPoly:
    def __init__(self, Ndegree):
        self.Ndegree = Ndegree
        # YOUR CODE HERE

    def fit(self, X, y):
        # construct the design matrix and determine beta
        # YOUR CODE HERE

    def predict(self, X):
        # YOUR CODE HERE

    def plot(self, ax, X):
        # YOUR CODE HERE
```

Fit a polynomial of degree two to the data from Task 1 and add the fitted curve to the plot. Check your implementation against

```
Xaug = [[ 1, -2,  4],
        [ 1,  0,  0],
        [ 1,  1,  1],
        [ 1,  2,  4],
        [ 1,  4, 16]]

beta = [-0.5, -1.8, 0.8]
```

Why does the fitted curve reproduce the sampled function exactly?

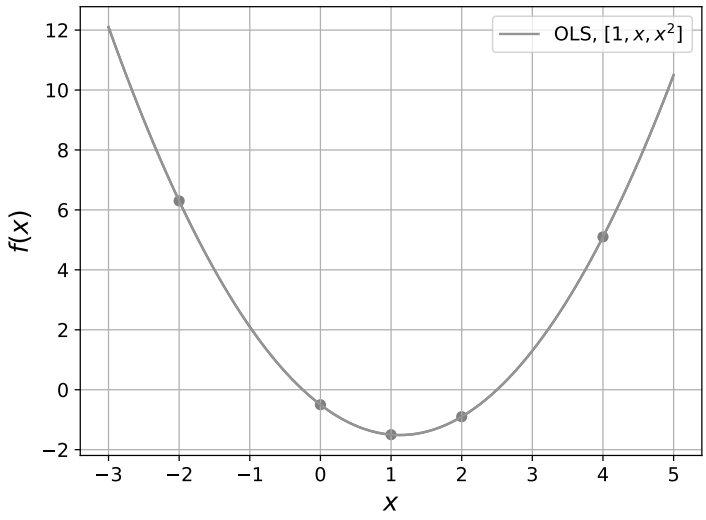


Figure 2: The quadratic OLS fit to the sampled data.

Task 3: fit noisy data

Add independent Gaussian noise to the sampled function values. For example, you may use a standard deviation of 0.2. Fit a polynomial of degree two to the noisy data and plot the original function, the noisy data, and the fitted curve in the same figure.

Repeat the calculation for several noise realizations. How do the fitted coefficients change? Why does the fitted curve no longer pass through every data point?

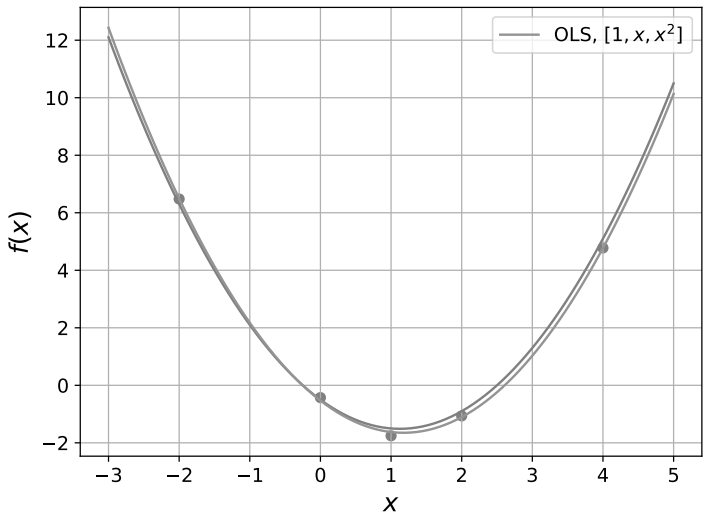


Figure 3: A quadratic OLS fit after adding noise to the sampled values.

Task 4: sample a double-well potential

Construct a one-dimensional double-well potential. You may reuse a potential and a molecular-dynamics implementation from an earlier exercise. Run a molecular-dynamics simulation and collect a sample of positions from the trajectory. Plot the potential and, if useful, a histogram representing the sampled position distribution.

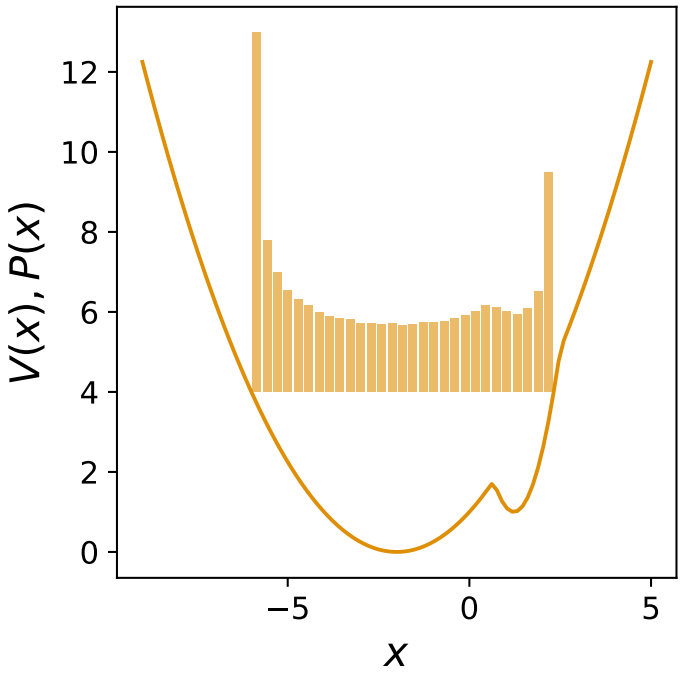


Figure 4: A double-well potential and positions sampled by molecular dynamics.

Select eight positions from the trajectory and evaluate the potential energy at these positions. Include positions from different regions of the sampled trajectory so that the resulting data are suitable for fitting. If you prefer not to repeat the molecular-dynamics calculation, you may instead choose eight positions directly. Plot the potential and the selected data points.

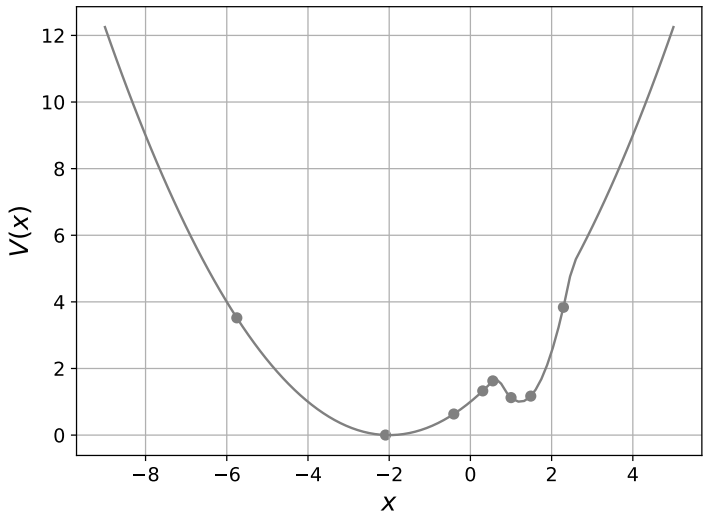


Figure 5: The double-well potential and the eight data points selected for fitting.

Task 5: fit the double-well data with a parabola

Use your OLS implementation to fit a polynomial of degree two to the selected double-well data. Plot the true potential, the selected data points, and the fitted curve in the same figure.

Discuss the result. Can a single parabola describe both wells and the barrier between them? Which aspects of the fitted curve are determined by the locations of the selected data points?

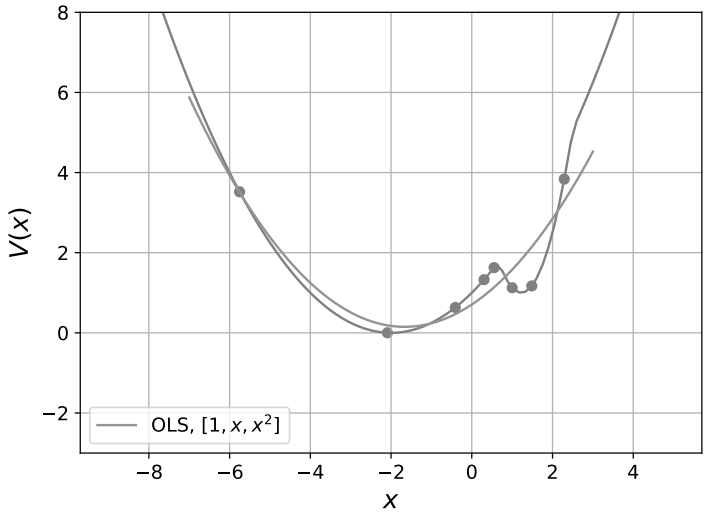


Figure 6: A quadratic OLS fit to data sampled from the double-well potential.

Task 6: vary the polynomial degree

Fit the same eight double-well data points with polynomials of degrees one through eight. Show the fits in a 2×4 arrangement of subplots. In every subplot, include the true potential and the selected data points, and use the same axis limits so that the fits can be compared directly.

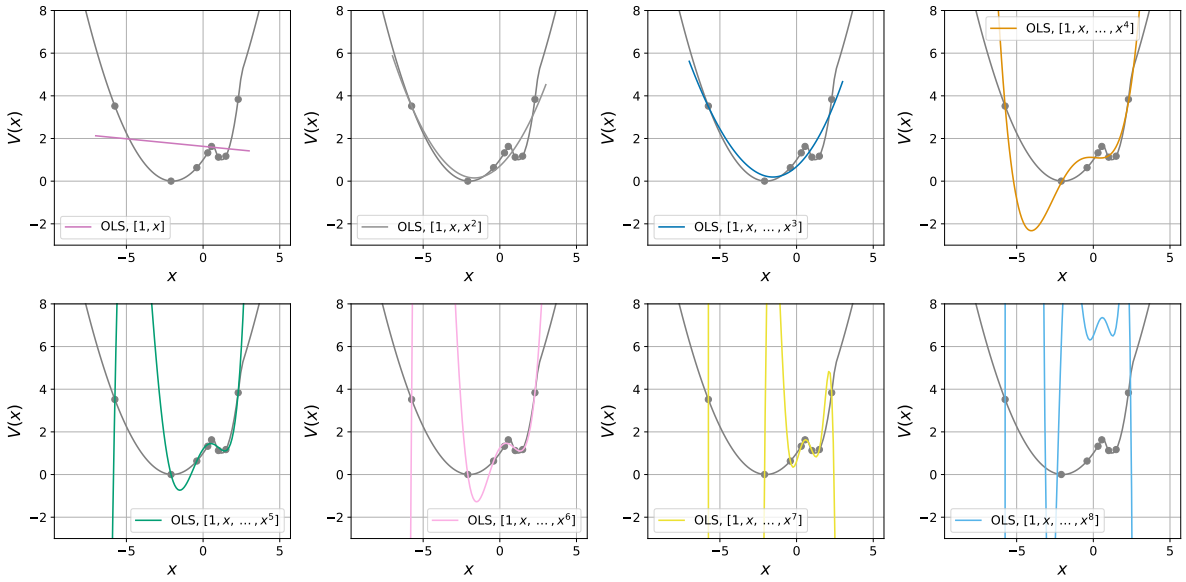


Figure 7: Polynomial OLS fits of increasing degree to the same eight data points.

For a polynomial of degree p, how many fitting parameters are there? Relate this number to the number of data points and explain what happens to $\mathbf{X}_{\text{aug}}^{\mathsf{T}}\mathbf{X}_{\text{aug}}$ when there are more parameters than data points. Comment also on the degree for which the number of parameters equals the number of data points. Does passing exactly through all sampled points imply that the model gives a useful representation of the potential between, or outside, those points?

Exercise W08B Ridge regression

In Exercise W08A, you observed that polynomial ordinary least squares fits can become unstable when the polynomial degree is increased. In particular, the matrix $\mathbf{X}_{\text{aug}}^{\mathsf{T}}\mathbf{X}_{\text{aug}}$ becomes singular when the model has more parameters than there are data points. In this exercise you will use ridge regression to regularize the polynomial fit.

For the polynomial feature vector

$$\boldsymbol{\phi}(x) = \begin{bmatrix} 1 & x & \cdots & x^p \end{bmatrix}^{\mathsf{T}},$$

the rows of $\mathbf{X}_{\text{aug}}$ contain the feature vectors of the sampled positions. Ridge regression determines the coefficients by minimizing

$$\left\|\boldsymbol{y} - \mathbf{X}_{\text{aug}}\boldsymbol{\beta}\right\|^2 + \lambda \left\|\boldsymbol{\beta}\right\|^2,$$

where $\lambda > 0$ is the regularization parameter. The solution is

$$\boldsymbol{\beta} = \left(\mathbf{X}_{\text{aug}}^{\mathsf{T}}\mathbf{X}_{\text{aug}} + \lambda\mathbf{I}\right)^{-1}\mathbf{X}_{\text{aug}}^{\mathsf{T}}\boldsymbol{y}.$$

Use the eight positions selected from the double-well potential in Exercise W08A throughout this exercise.

Task 1: implement ridge regression

Implement polynomial ridge regression in a class with `fit`, `predict`, and `plot` methods. The polynomial degree and $\log_{10}(\lambda)$ should be supplied when the class is instantiated. You may reuse and extend the polynomial OLS implementation from Exercise W08A.

Fit a polynomial of degree five to the eight double-well data points using ridge regression with $\lambda = 1$. In the same figure, show the double-well potential, the data points, the ridge-regression fit, and the corresponding OLS fit. Use the same polynomial features for the two fits.

How do the two fits differ, both close to the sampled points and away from them? Compare the sizes of the fitted coefficients. Explain how the regularization term influences the coefficients.

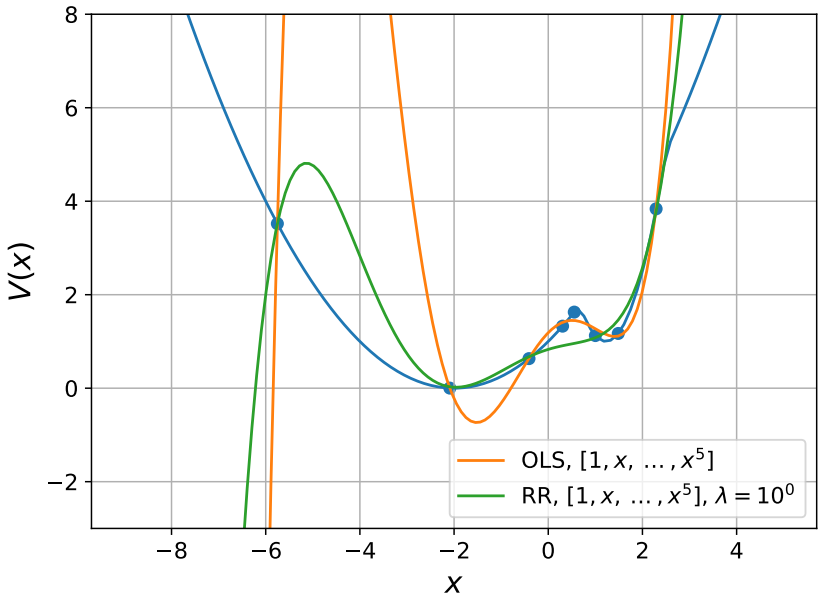


Figure 8: Polynomial fits obtained with ordinary least squares and ridge regression for the same double-well data.

Task 2: vary the polynomial degree

Fit the same eight data points with ridge-regression models of degrees one through eight using $\lambda = 10$. Show the fits in a 2×4 arrangement of subplots. In every subplot, include the true potential and the data points, and use the same axis limits so that the fits can be compared directly.

Compare the result with the OLS fits from Exercise W08A. In particular, discuss the fit for which the number of parameters exceeds the number of data points. Why can the ridge-regression coefficients still be determined? Explain the role of the term $\lambda \mathbf{I}$ in the matrix that is inverted.

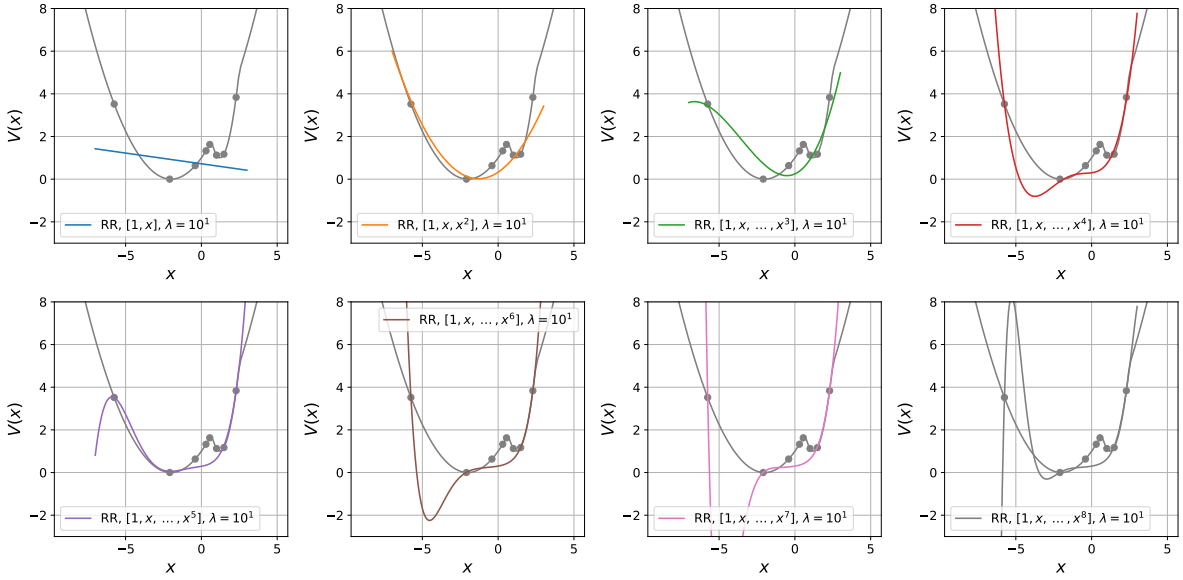


Figure 9: Ridge-regression fits of increasing polynomial degree to the same eight data points.

Task 3: vary the regularization parameter

Use a polynomial of degree six and fit the same data for

$$\log_{10}(\lambda) = -2, -1, 0, 1, 2, 3, 4, 5.$$

Show the eight fits in a 2×4 arrangement of subplots. Use the same axis limits in all subplots and include the true potential and the data points.

Describe the transition from weak to strong regularization. What happens to the fitted coefficients and to the predicted function as λ is increased? What are the limiting predictions for $\lambda \to 0$ and $\lambda \to \infty$? Identify a range of λ values that gives a reasonable fit to the sampled potential.

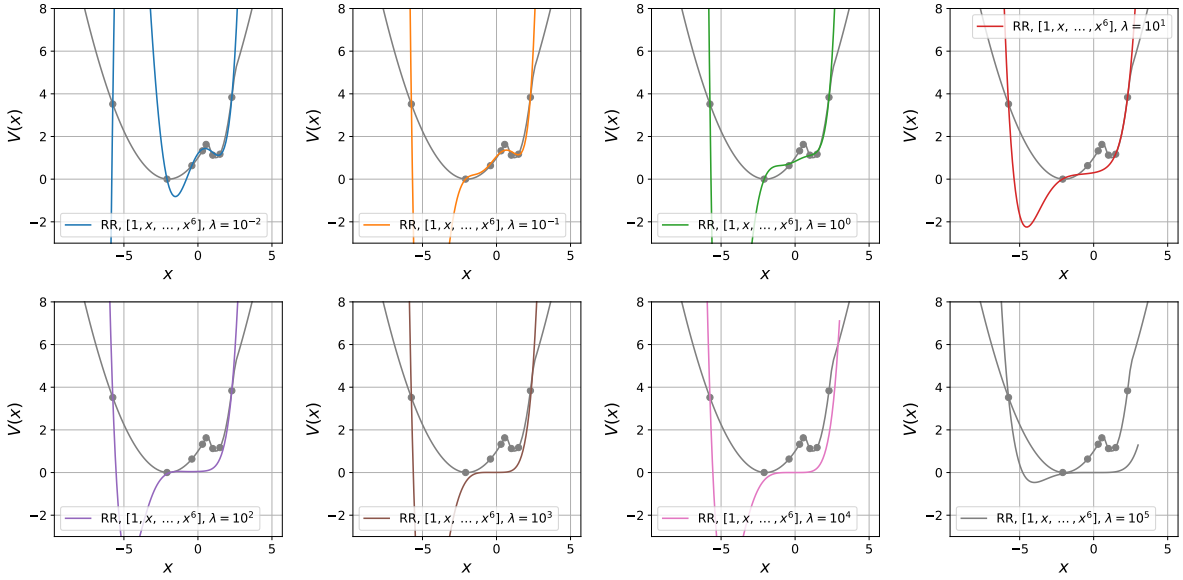


Figure 10: Degree-six ridge-regression fits for increasing values of the regularization parameter.

Task 4: include a prior model

Extend your ridge-regression implementation so that it can include a prior model $f_{\text{prior}}(x)$. Use a first-degree polynomial fitted to the data with OLS as the prior model. The ridge-regression correction should minimize

$$\sum_i \left[y_i - f_{\text{prior}}(x_i) - \sum_j \beta_j \phi_j(x_i) \right]^2 + \lambda \sum_j \beta_j^2.$$

Thus, fit the ridge-regression model to the residual values

$$y_i - f_{\text{prior}}(x_i),$$

and add the prior prediction again when making a prediction:

$$f(x) = f_{\text{prior}}(x) + \sum_j \beta_j \phi_j(x).$$

Use a degree-six ridge-regression correction and repeat the sequence of regularization parameters from Task 3. Show the prior model, the true potential, the data points, and the combined prediction in each subplot.

Investigate the limit of large regularization. Explain why the combined model approaches the prior model rather than zero when λ becomes large. What advantage can a physically or statistically meaningful prior provide when the data are sparse?

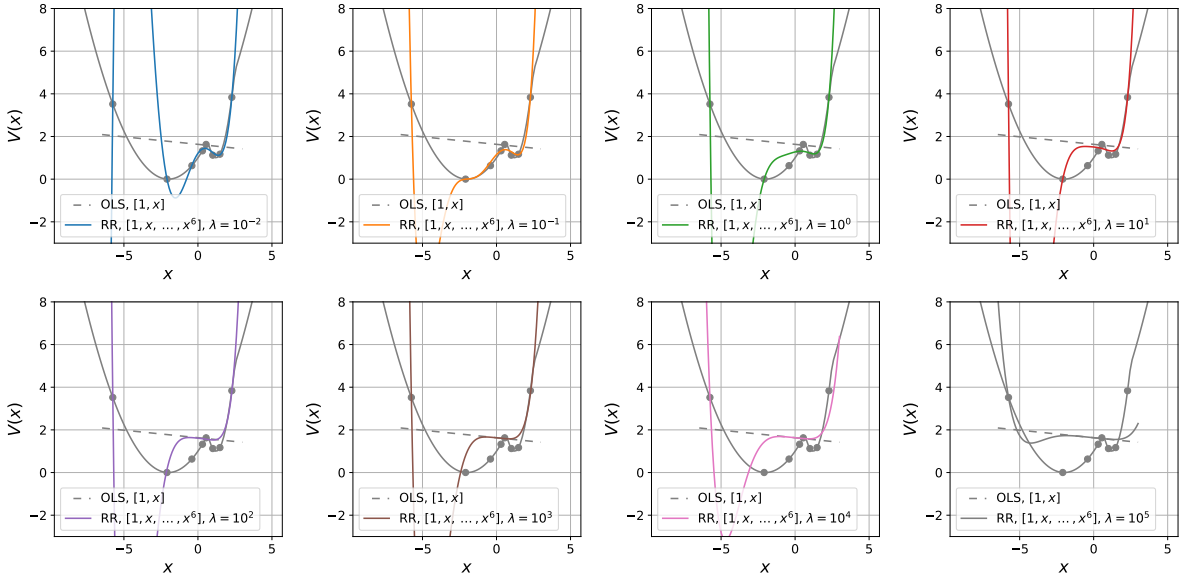


Figure 11: Degree-six ridge-regression corrections to a first-degree OLS prior for increasing values of the regularization parameter.

Exercise W08C Kernel ridge regression with a Gaussian kernel

In this exercise you will implement kernel ridge regression using the Gaussian kernel. You will first evaluate the kernel for individual points and construct a kernel matrix. You will then use the kernel to fit one-dimensional data and investigate the roles of the kernel width, a prior expectation, and regularization.

The Gaussian kernel is

$$K(\boldsymbol{x}_1, \boldsymbol{x}_2) = \exp\left(-\frac{\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|^2}{2\sigma^2}\right),$$

where σ is the kernel width. For training inputs x_i, kernel ridge regression predicts

$$\hat{y}(x) = \sum_{j=1}^{N} K(x, x_j)\alpha_j,$$

where the coefficients are obtained from

$$\boldsymbol{\alpha} = (\mathbf{K} + \lambda\mathbf{I})^{-1}\boldsymbol{y}, \qquad K_{ij} = K(x_i, x_j).$$

Task 1: implement the Gaussian kernel

Implement the Gaussian kernel and evaluate $K(1, 2)$ using $\sigma = 2$. Check that you obtain approximately

```
0.8824969025845955
```

Task 2: evaluate the kernel between one point and several points

Introduce the three data points

```
X = np.array([1, 2, 5])
```

and evaluate the kernel between $x = 1$ and every point in `X`. Arrange your implementation so that it can evaluate several query points at once. Check that the result is

```
[[1.    0.882 0.135]]
```

What determines how quickly the kernel value decreases as the distance between two points increases?

Task 3: construct the kernel matrix

Evaluate the kernel matrix $\mathbf{K}$ with elements

$$K_{ij} = K(x_i, x_j),$$

where both x_i and x_j run over all points in `X`. Check that you obtain

```
[[1.    0.882 0.135]
 [0.882 1.    0.325]
 [0.135 0.325 1.   ]]
```

Why are all diagonal elements equal to one? Why is the matrix symmetric?

Task 4: fit data with kernel ridge regression

Use the following data:

```
xs = np.array([ 1.00073656,  2.29030314, -2.09037468,
               -5.75259232,  0.30391792,  0.55216248,
                1.48650712, -0.40884007])
ys = np.array([1.12573393e+00, 3.83712913e+00,
               2.04189572e-03, 3.52048728e+00,
               1.32700944e+00, 1.62773413e+00,
               1.17096493e+00, 6.32947483e-01])
```

Fit the data using kernel ridge regression with the Gaussian kernel. Use a small positive regularization parameter rather than inverting $\mathbf{K}$ directly. Evaluate the prediction on a dense grid and plot the data together with the predicted curve.

Repeat the fit for $\sigma = 0.1$, 0.5, and 2. Show the three results next to each other using identical axis limits. Explain how σ controls the smoothness and spatial extent of the prediction.

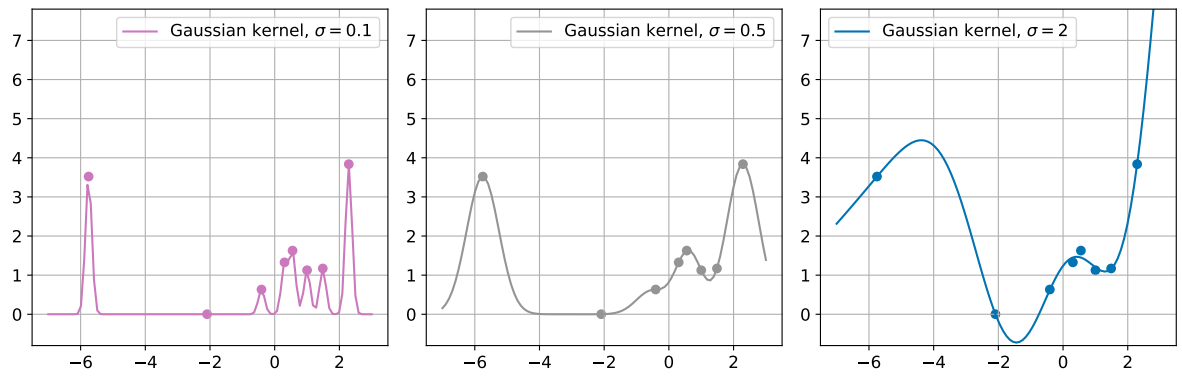


Figure 12: Kernel ridge regression fits obtained with three Gaussian kernel widths.

Task 5: introduce a constant prior

Far from all training points, every Gaussian kernel value approaches zero. Consequently, the prediction above approaches zero as well. Modify the method so that the prediction is made relative to a constant prior expectation y_0. Fit the deviations $y_i - y_0$ and add y_0 to the prediction:

$$\boldsymbol{\alpha} = (\mathbf{K} + \lambda \mathbf{I})^{-1}(\mathbf{y} - y_0 \mathbf{1}), \qquad \hat{y}(x) = y_0 + \sum_{j=1}^{N} K(x, x_j)\alpha_j .$$

Using $\sigma = 0.5$, compare the predictions for $y_0 = 1$, 2.5, and 5. Include each prior as a horizontal dashed line. What value does each prediction approach far from the training data?

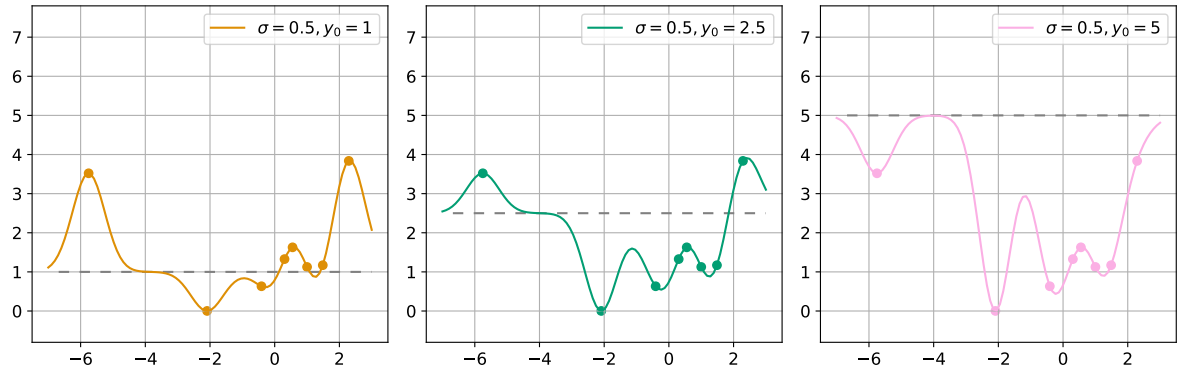


Figure 13: Kernel ridge regression relative to three constant prior expectations.

Task 6: investigate a nonconstant prior and regularization

Replace the constant prior by a function $f_0(x)$. Fit the residuals $y_i - f_0(x_i)$ and use

$$\hat{y}(x) = f_0(x) + \sum_{j=1}^{N} K(x, x_j)\alpha_j.$$

Choose a simple nonconstant prior, for example a parabola, and plot it together with the data and the resulting kernel ridge regression prediction.

Keep the kernel width fixed and compare several values of the regularization parameter, for example $\lambda = 10^{-4}$, 10^{-2}, and 1. Show the results next to each other using identical axis limits. Discuss how increasing λ changes the fitted corrections to the prior. In which limit does the prediction remain close to the prior, and in which limit does it follow the training data most closely?

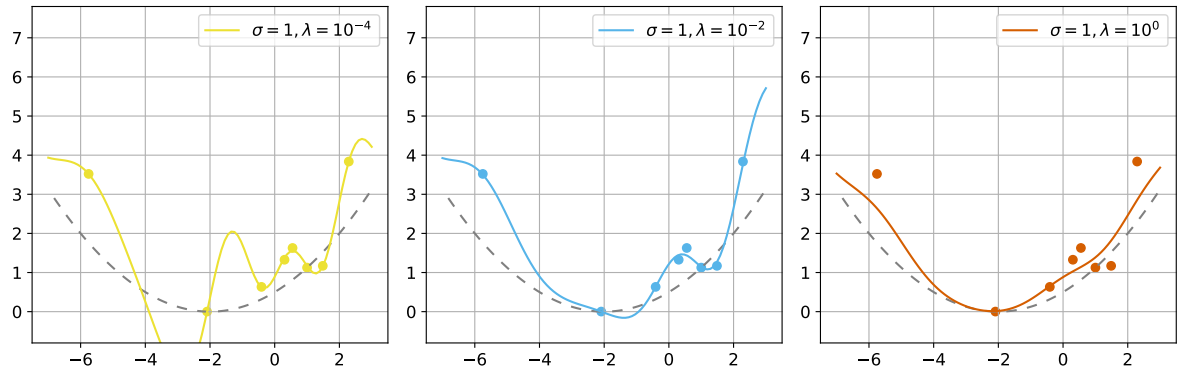


Figure 14: Kernel ridge regression relative to a nonconstant prior for three regularization strengths.

Exercise W08D Implicit feature mappings of kernels

Kernel ridge regression uses a kernel $K(x_1, x_2)$ to evaluate inner products between feature vectors without constructing the feature vectors explicitly. In this exercise you will construct explicit feature mappings for the second-degree polynomial kernel and the radial basis function (RBF) kernel, and verify numerically that their inner products reproduce the corresponding kernels.

Task 1: the feature mapping of a polynomial kernel

Consider the second-degree polynomial kernel

$$K(x_1, x_2) = (1 + x_1 x_2)^2.$$

It corresponds to the feature mapping

$$\boldsymbol{\phi}(x) = \begin{bmatrix} 1 & \sqrt{2}x & x^2 \end{bmatrix}^\mathsf{T}.$$

Set $x_1 = 2$ and let x_2 vary over the interval $-5 < x_2 < 5$. Evaluate $\boldsymbol{\phi}(x_1)^\mathsf{T}\boldsymbol{\phi}(x_2)$ and $K(x_1, x_2)$ directly, and plot both quantities as functions of x_2 in the same figure. Verify both algebraically and numerically that they are identical.

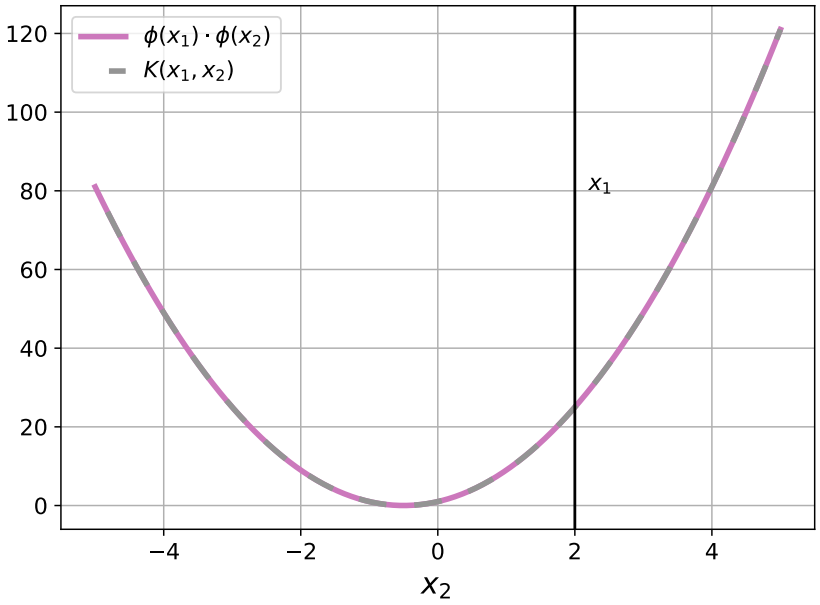


Figure 15: Comparison of the explicit feature-vector inner product and the second-degree polynomial kernel for fixed $x_1 = 2$.

Task 2: parity plot for the polynomial kernel

Draw 1000 pairs (x_1, x_2) independently from the uniform distribution on $[-5, 5)$. For every pair, calculate $\boldsymbol{\phi}(x_1)^\mathsf{T}\boldsymbol{\phi}(x_2)$ and $K(x_1, x_2)$. Make a parity plot with the feature-vector inner product

on the horizontal axis and the kernel value on the vertical axis. What relation should the points follow?

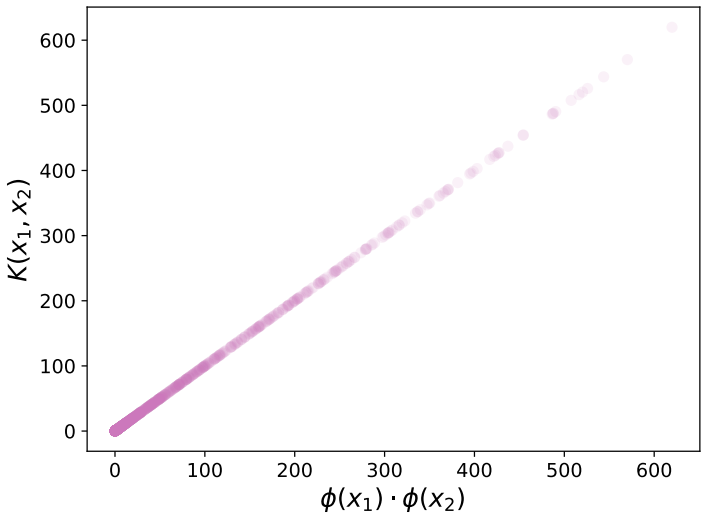


Figure 16: Parity plot comparing the explicit feature-vector inner products with the polynomial-kernel values.

Task 3: a discrete feature mapping of the RBF kernel

Consider the RBF kernel

$$K(x_1, x_2) = \exp\left[-\frac{(x_1 - x_2)^2}{2\sigma^2}\right].$$

An explicit, discretized feature mapping can be constructed from

$$\phi_i(x) = A\frac{1}{\sqrt{2\pi}\sigma_\phi}\exp\left[-\frac{(x - x_i)^2}{2\sigma_\phi^2}\right]\sqrt{\Delta x},$$

where

$$\sigma_\phi = \frac{\sigma}{\sqrt{2}}, \qquad A = \sqrt{\sqrt{2\pi}\sigma}, \qquad x_{i+1} = x_i + \Delta x.$$

The feature vector is

$$\mathbf{\Phi}(x) = \begin{bmatrix}\cdots & \phi_{i-1}(x) & \phi_i(x) & \phi_{i+1}(x) & \cdots\end{bmatrix}^\mathsf{T}.$$

Implement this mapping as a class named `DiscreteRBF`. Let the class construct the grid points x_i and return $\mathbf{\Phi}(x)$ when called. Use $\sigma = 0.1$ and choose the range of grid points wide enough that truncation of the Gaussian functions is negligible for all values of x used below. You may reuse your descriptor base class from an earlier exercise if convenient.

For $x_1 = 0.25$ and $x_2 = 0.50$, evaluate

$$\mathbf{\Phi}(x_1)^\mathsf{T}\mathbf{\Phi}(x_1), \qquad \mathbf{\Phi}(x_2)^\mathsf{T}\mathbf{\Phi}(x_2), \qquad \mathbf{\Phi}(x_1)^\mathsf{T}\mathbf{\Phi}(x_2)$$

for

$$\Delta x = 0.1,\ 0.02,\ 0.04,\ 0.008,\ 0.0016,\ 0.00032,\ \text{and}\ 0.000064.$$

Check that the first two inner products approach unity as the discretization is refined. Compare the third inner product with $K(x_1, x_2)$. Explain why a coarse grid can give a less accurate result and why the feature vector becomes higher-dimensional as Δx is reduced.

Task 4: visualize the discrete RBF feature vectors

For $\Delta x = 0.20$, 0.15, 0.10, and 0.05, plot the components of $\mathbf{\Phi}(0.50)$ and $\mathbf{\Phi}(0.25)$ as functions of their grid centers x_i. Place the four discretizations in a 1×4 arrangement of subplots and use common axis limits to facilitate comparison.

How do the sampled Gaussian shapes change as Δx is reduced? Relate the number of non-negligible components to the resolution of the discretization.

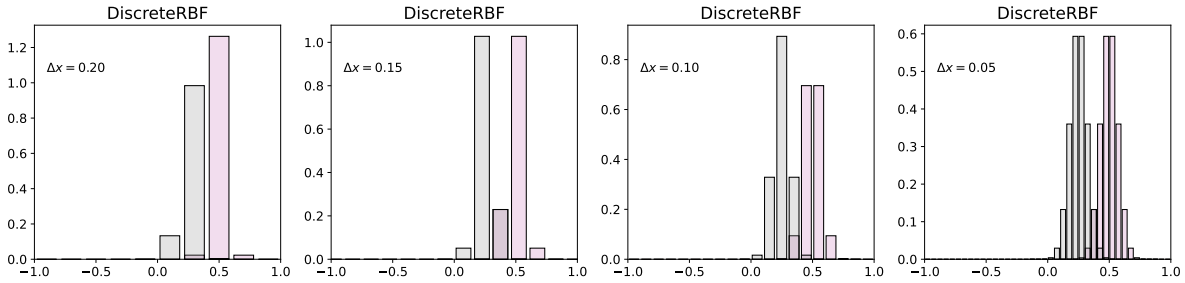


Figure 17: Discrete RBF feature vectors for two input values and four grid spacings.

Task 5: recover the RBF kernel from feature-vector inner products

Set $x_1 = 2$ and let x_2 vary over the interval $1 < x_2 < 3$. For each of the four grid spacings from Task 4, evaluate $\mathbf{\Phi}(x_1)^{\mathsf{T}}\mathbf{\Phi}(x_2)$ and $K(x_1, x_2)$. Plot the two quantities as functions of x_2 in a 1×4 arrangement of subplots.

How does the agreement depend on Δx? Explain why the two quantities coincide in the limit $\Delta x \to 0$.

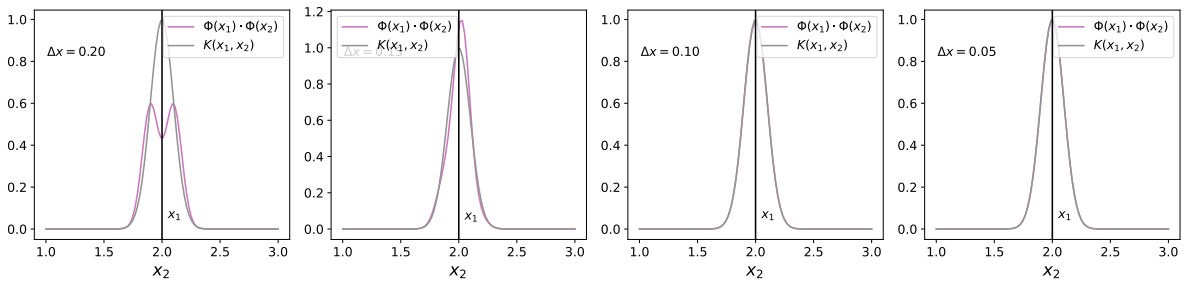


Figure 18: Comparison of the discrete feature-vector inner products with the RBF kernel for four grid spacings.

Task 6: parity plots for the RBF kernel

Draw 1000 pairs (x_1, x_2) independently from the uniform distribution on $[-5, 5)$. For each of the four grid spacings from Task 4, make a parity plot of $K(x_1, x_2)$ as a function of $\mathbf{\Phi}(x_1)^\mathsf{T}\mathbf{\Phi}(x_2)$. Use equal aspect ratios and the same axis limits in all subplots.

Discuss the deviations from the parity line for the coarsest discretizations. What does this calculation demonstrate about the implicit feature mapping used by kernel ridge regression?

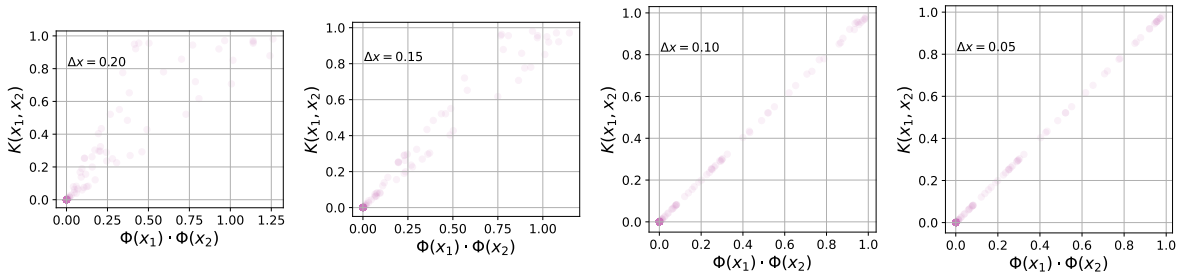


Figure 19: Parity plots comparing discrete feature-vector inner products with RBF-kernel values for four grid spacings.

Exercise W08E Kernel ridge regression with atomic descriptors

In this exercise you will investigate how the representation of an atomic configuration influences a kernel ridge regression model. The data have only one independent degree of freedom: the x-coordinate of one atom in an atomic cluster. You will compare regression based directly on this coordinate with regression based on two higher-dimensional descriptors of the complete atomic configuration.

The Gaussian kernel between two descriptor vectors $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ is

$$K_{ij} = \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{2\sigma^2}\right).$$

For training targets collected in the vector $\boldsymbol{y}$, kernel ridge regression determines the coefficient vector

$$\boldsymbol{\alpha} = (\mathbf{K} + \lambda\mathbf{I})^{-1}\,\boldsymbol{y}.$$

The prediction for a new descriptor vector $\boldsymbol{x}$ is

$$\hat{E}(\boldsymbol{x}) = \sum_i \alpha_i K(\boldsymbol{x}, \boldsymbol{x}_i).$$

Task 1: construct the atomic-cluster data

Use the `Morse` calculator and the `AtomicCluster` class from Week 04 to construct an atomic cluster in which all atoms except one are fixed. Constrain the movable atom to positions $(x, 0)$ and use the following eight sampled x-coordinates:

$$x_i = -2.0, -1.7, -1.6, -1.4, -1.2, -0.8, -0.5, 0.0.$$

For each value of x_i, calculate the potential energy E_i and visualize the corresponding atomic configuration. Show the eight configurations in a 2×4 arrangement of subplots, using the same axis limits and equal aspect ratios in all subplots.

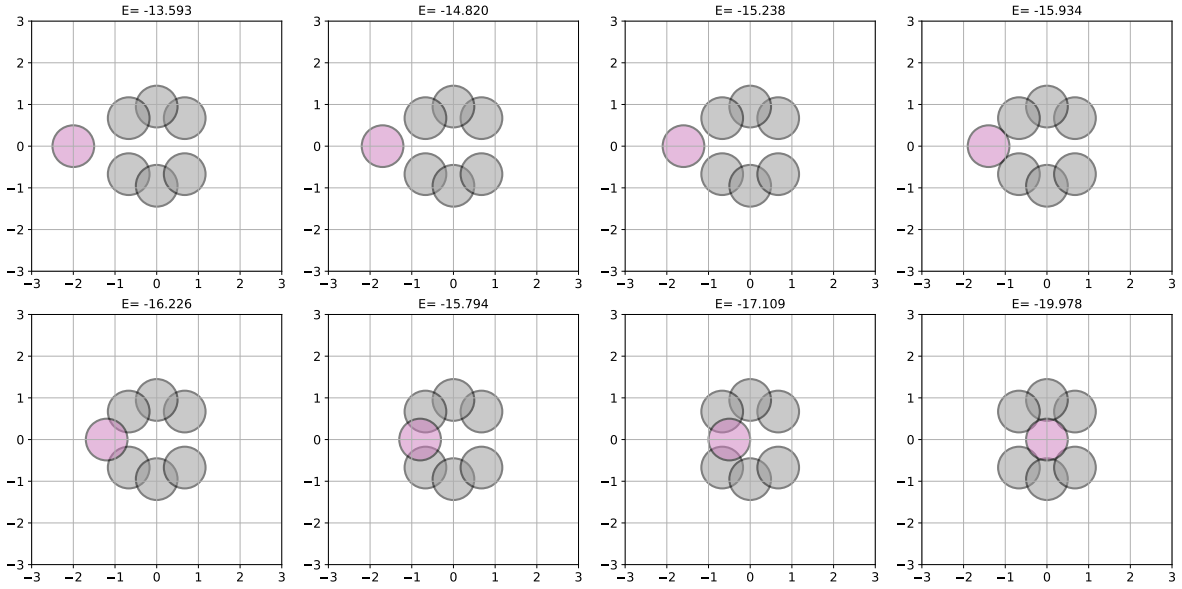


Figure 20: The eight atomic configurations used as training data. Only one atom changes position.

Evaluate the energy on a dense grid covering $-2 < x < 2$. Plot this energy curve and indicate the eight sampled configurations as points. This dense curve will serve as a reference when the regression models are assessed.

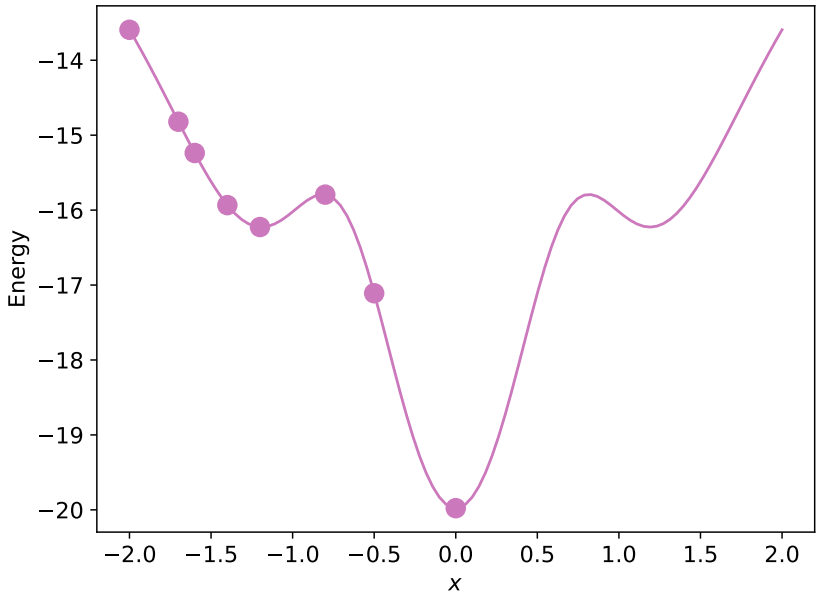


Figure 21: Potential energy as a function of the movable atom's x-coordinate, with the eight training points indicated.

Task 2: establish three representations of the data

Construct three datasets with the same target energies E_i but different representations of the input configurations:

$$(x_i, E_i),$$

$$\left(\boldsymbol{f}_{\text{PairDistances}}(x_i), E_i\right),$$

and

$$\left(\boldsymbol{f}_{\text{CoulombMatrixSpectrum}}(x_i), E_i\right).$$

The pair-distance descriptor is a histogram of all interatomic distances. Use distances from 0 to 7 and a bin width of 0.5, giving 14 descriptor components. The Coulomb-matrix-spectrum descriptor is the sorted spectrum of the matrix whose off-diagonal elements are the inverse interatomic distances. Set the diagonal distances to one before taking the inverse, as in your implementation from the earlier descriptor exercises.

Inspect the shapes and values of the three input matrices. How many features does each representation contain? Explain why changing one atomic coordinate can change several components of either atomic descriptor.

Task 3: fit without an atomic descriptor

Use the first dataset to make a kernel ridge regression fit with the Gaussian kernel. In this model, the prediction is based only on the scalar coordinate x. Use $\lambda = 10^{-4}$ and compare the results obtained with $\sigma = 1$ and $\sigma = 10$.

Plot the reference energy curve, the training points, and both kernel ridge regression predictions in the same figure. Discuss how the width of the kernel influences interpolation between the sampled points and extrapolation outside the sampled interval. Why can two models fitted to the same training data give very different predictions between and beyond those data?

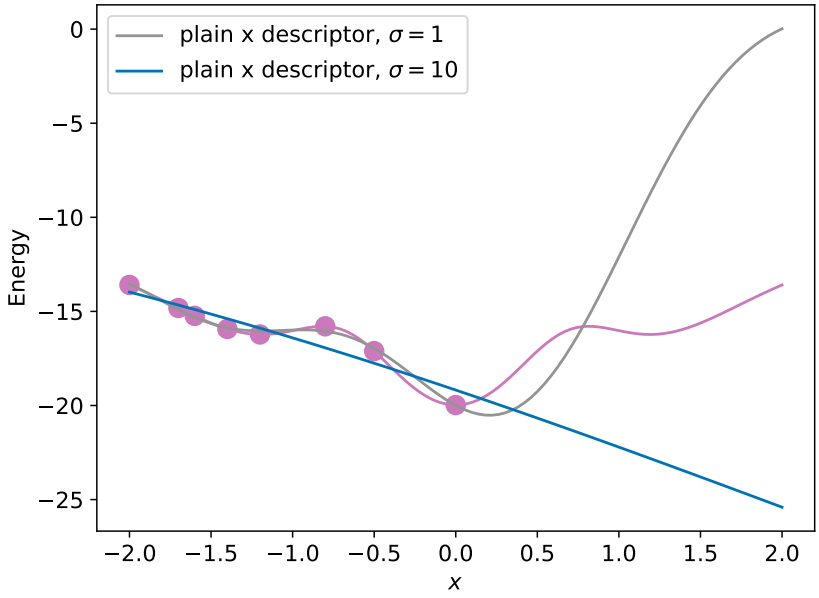


Figure 22: Kernel ridge regression based directly on the movable atom's x-coordinate for two Gaussian-kernel widths.

Task 4: fit using the pair-distance descriptor

Use the second dataset to make a kernel ridge regression fit based on the 14-dimensional pair-distance descriptor. Use a Gaussian kernel with $\sigma = 5$ and regularization parameter $\lambda = 0.01$.

For every point on the dense x-grid, construct the corresponding atomic configuration, evaluate its pair-distance descriptor, and use this descriptor to predict the energy. Plot the reference energy curve, the training points, and the prediction.

The Gaussian kernel now evaluates Euclidean distances in the 14-dimensional descriptor space rather than distances along the original x-coordinate. Discuss how this changes the similarities between configurations. Pay particular attention to any consequence of using a histogram descriptor, for which the components change discretely when an interatomic distance crosses a bin boundary.

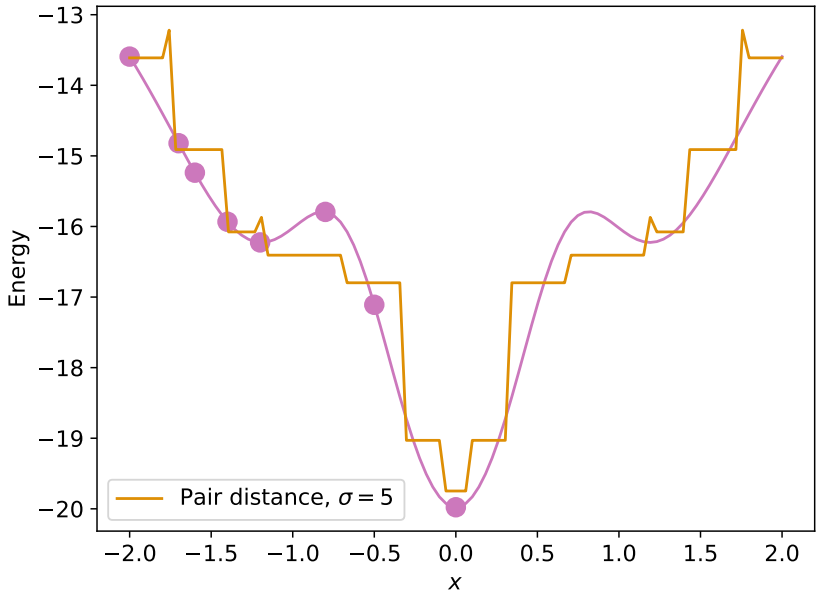


Figure 23: Kernel ridge regression based on the pair-distance descriptor.

Task 5: fit using the Coulomb-matrix-spectrum descriptor

Repeat Task 4 using the Coulomb-matrix-spectrum descriptor. Use a Gaussian kernel with $\sigma = 1$ and regularization parameter $\lambda = 0.01$. Construct the descriptor for every configuration on the dense x-grid and use it to predict the energy.

Plot the reference energy curve, the training points, and the prediction. Compare this fit with those based on the scalar coordinate and the pair-distance descriptor. Which representation gives the most useful notion of similarity between the atomic configurations in this example? Discuss the smoothness, interpolation, and extrapolation of the three models, and explain why a higher-dimensional descriptor is not automatically a better descriptor.

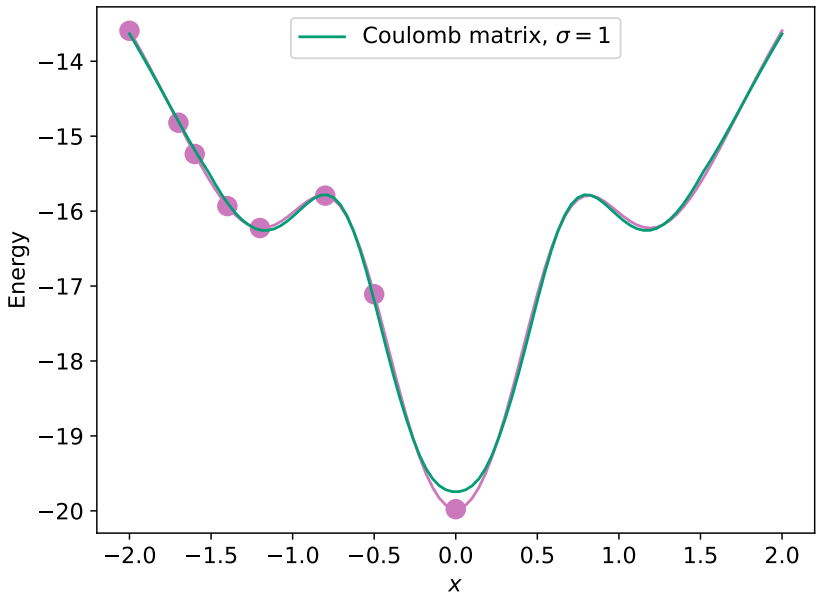


Figure 24: Kernel ridge regression based on the Coulomb-matrix-spectrum descriptor.